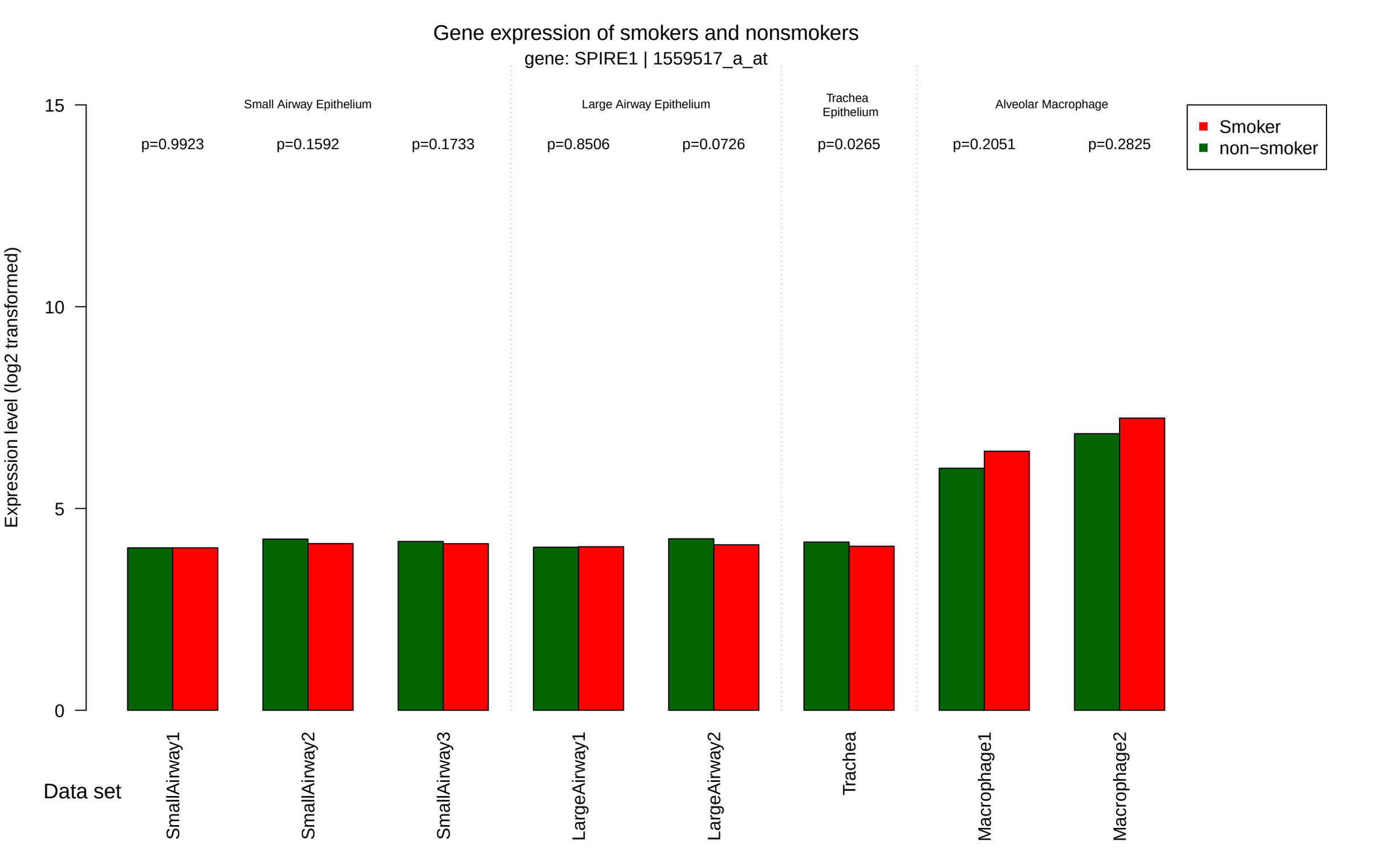
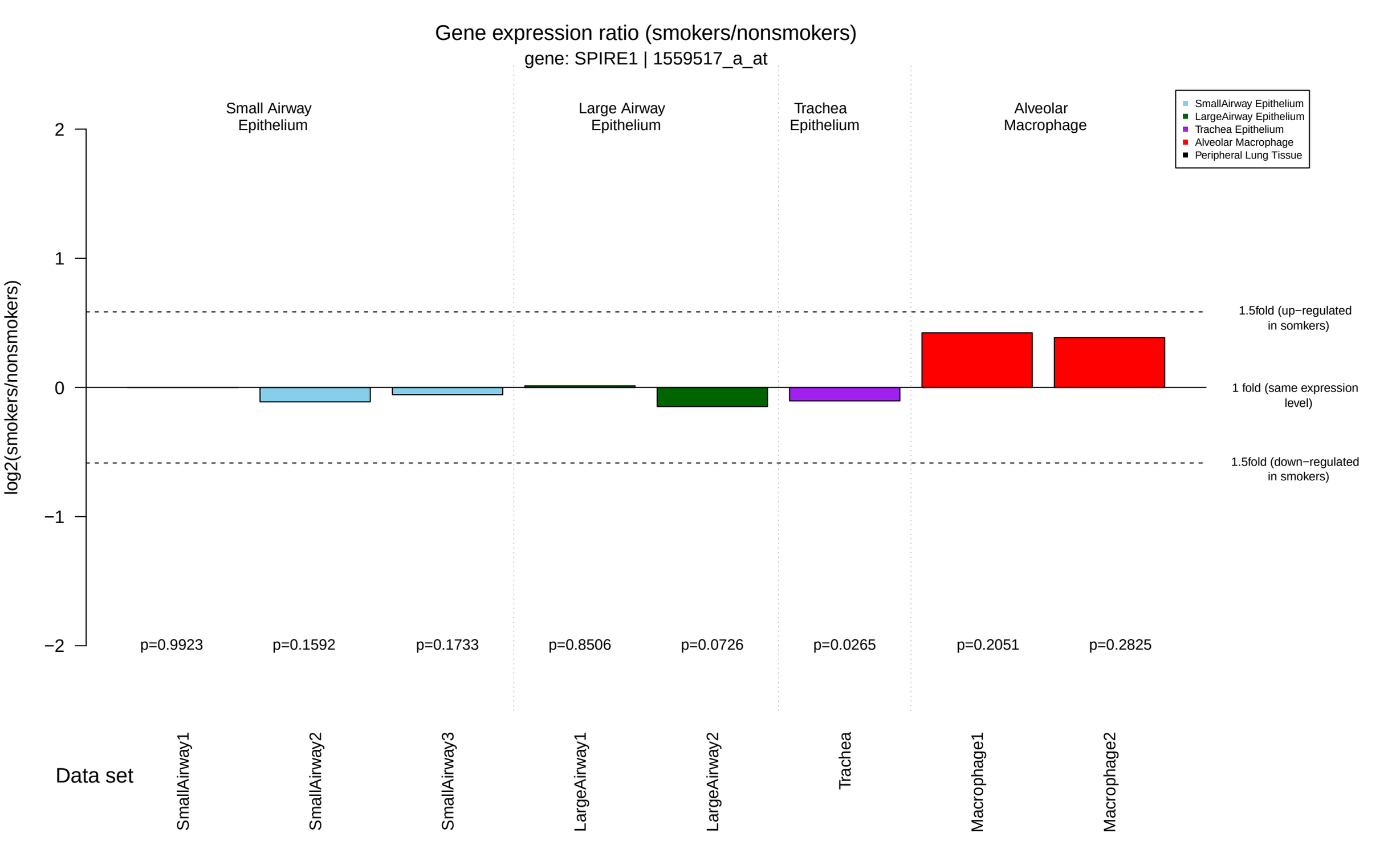
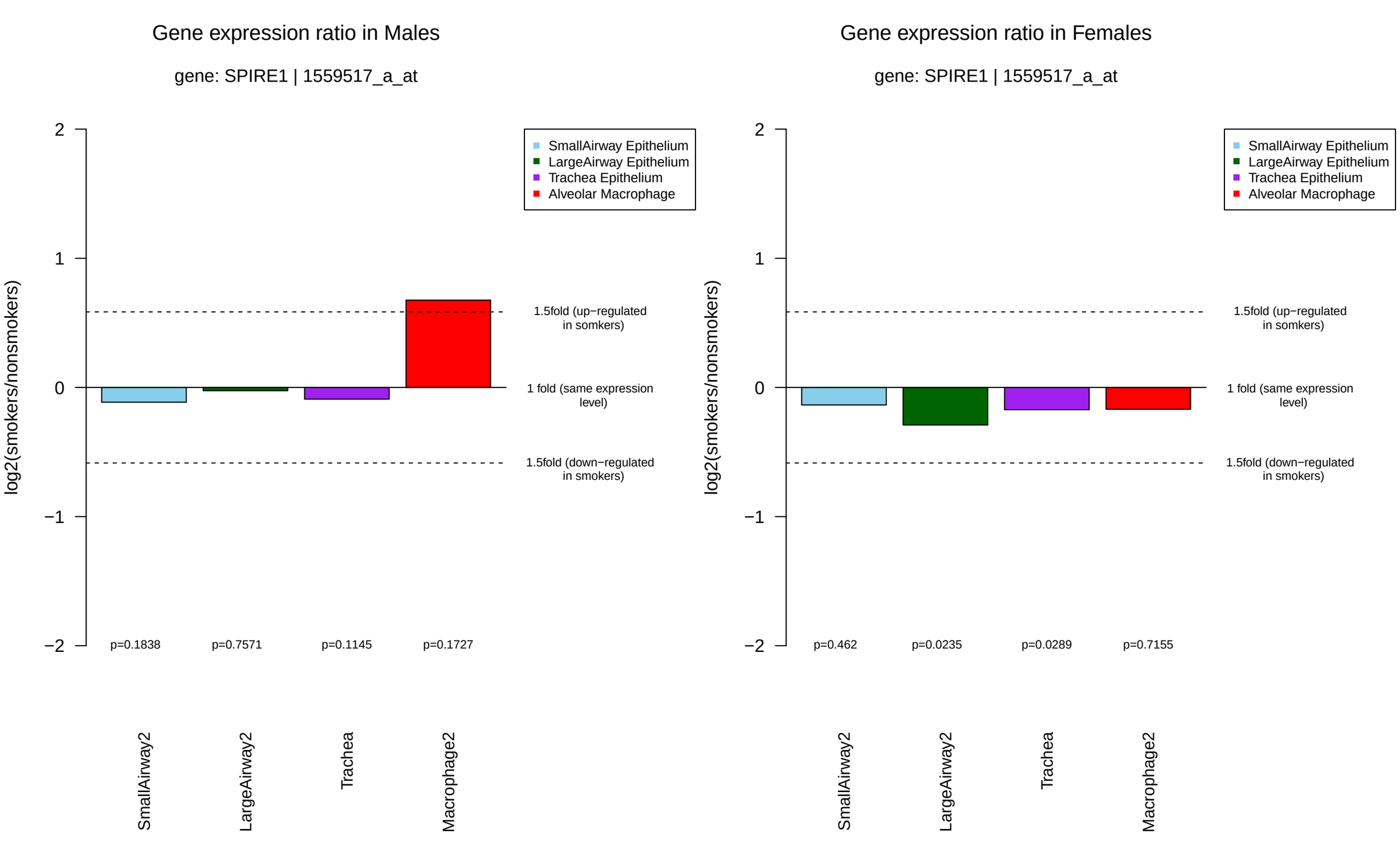
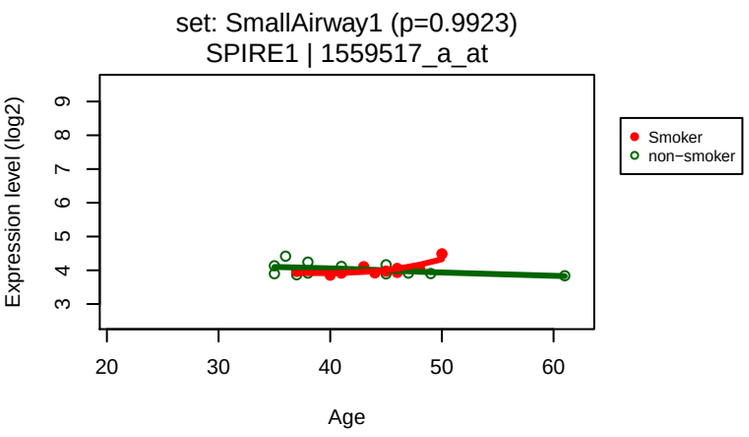
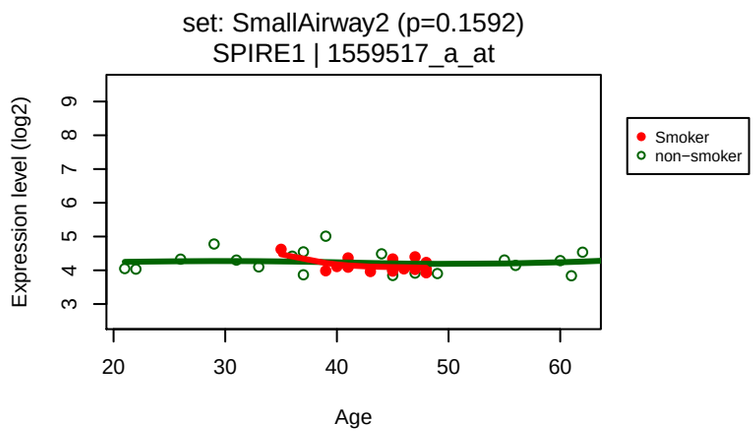
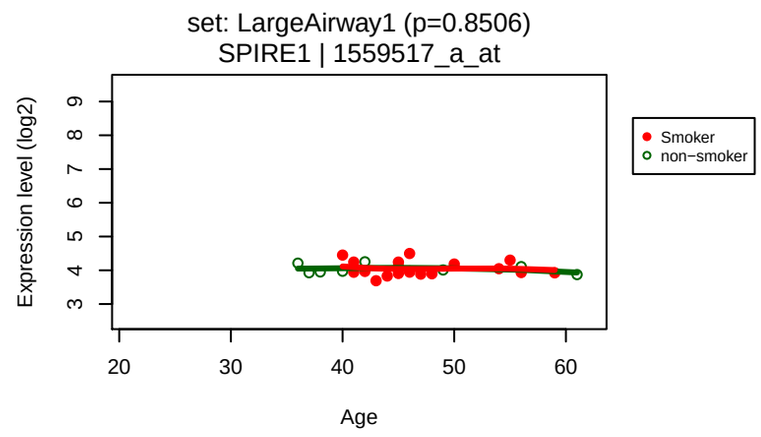

Gene expression of smokers and nonsmokers
gene: SPIRE1 | 1559517_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9923
p=0.1592
p=0.1733
p=0.8506
p=0.0726
p=0.0265
p=0.2051
p=0.2825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

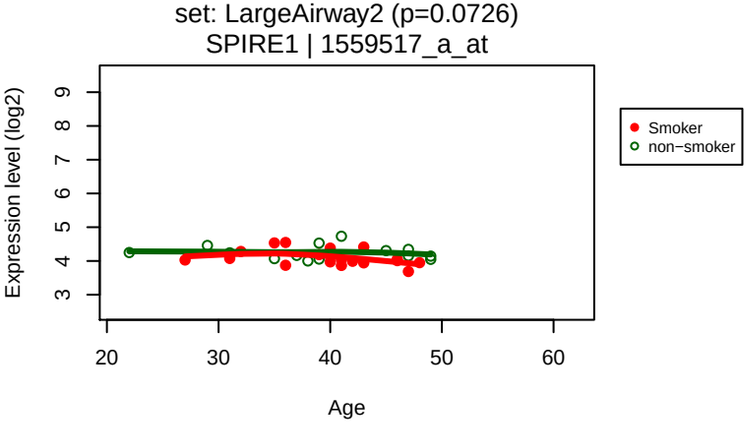

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE1 | 1559517_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9923
p=0.1592
p=0.1733
p=0.8506
p=0.0726
p=0.0265
p=0.2051
p=0.2825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

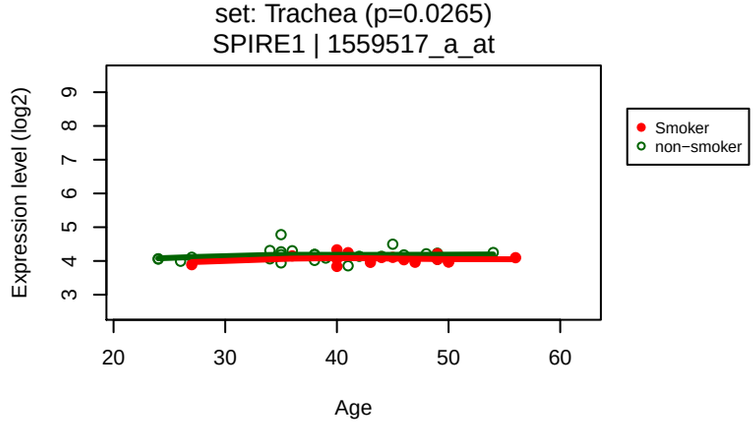

Gene expression ratio in Males
gene: SPIRE1 | 1559517_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1838
p=0.7571
p=0.1145
p=0.1727
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE1 | 1559517_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.462
p=0.0235
p=0.0289
p=0.7155
SmallAirway2
LargeAirway2
Trachea
Macrophage2

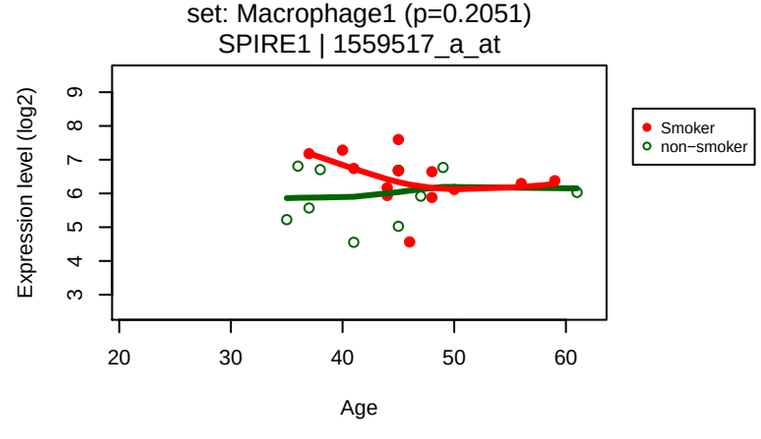

set: SmallAirway1 (p=0.9923)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker

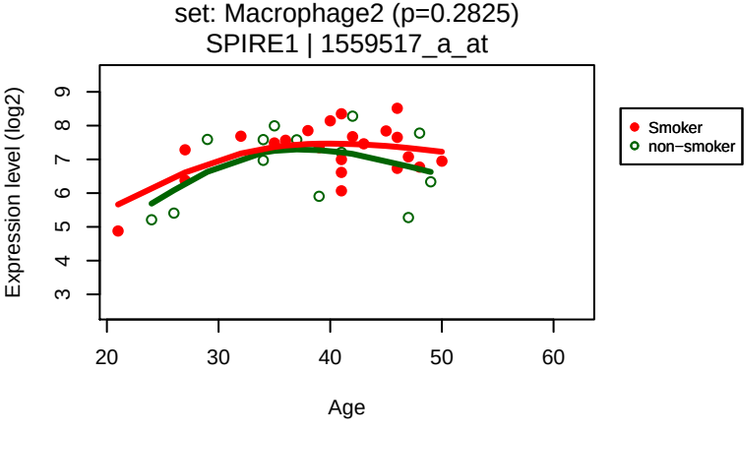

set: SmallAirway2 (p=0.1592)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker

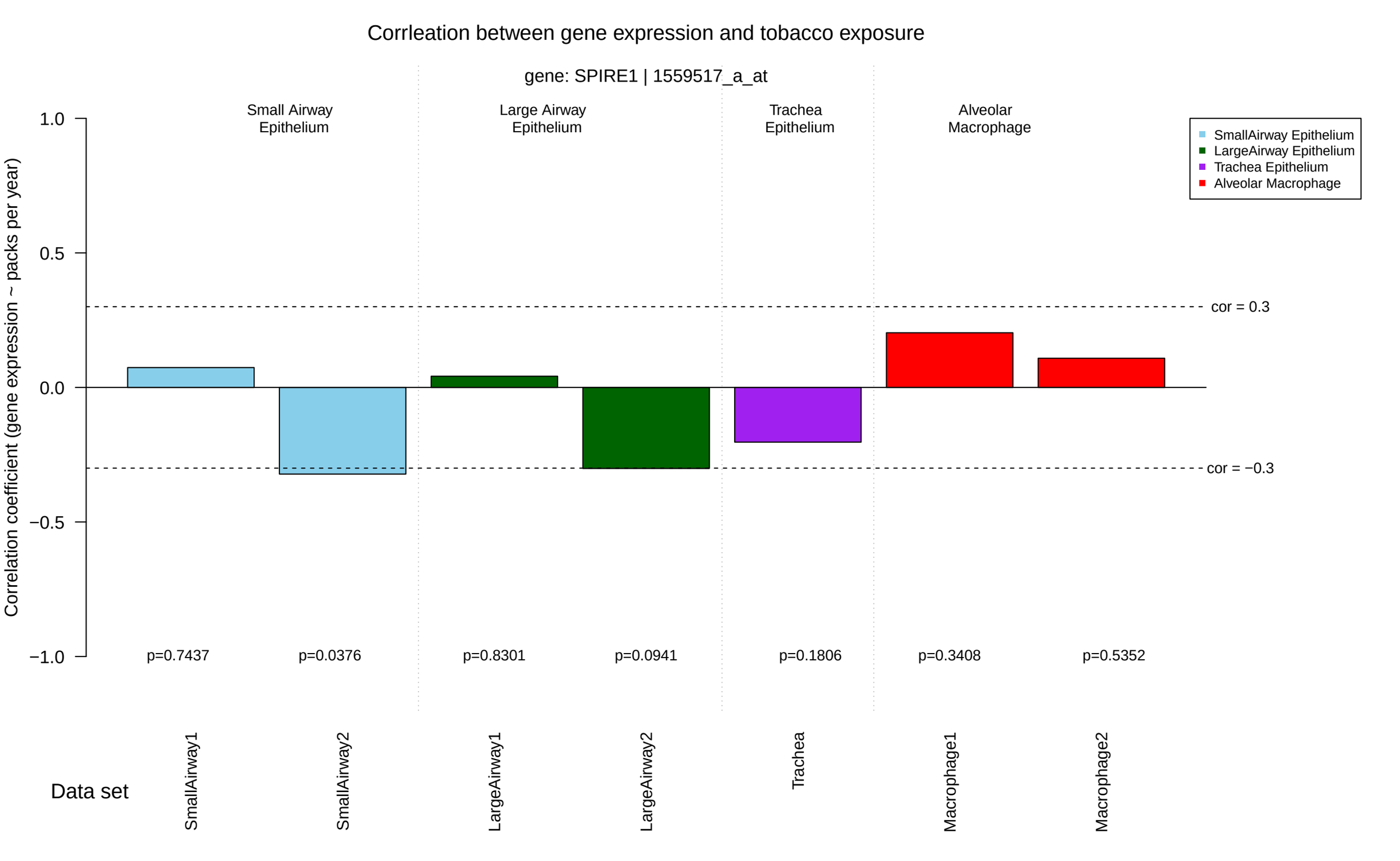

set: LargeAirway1 (p=0.8506)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0726)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0265)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2051)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2825)
SPIRE1 | 1559517_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE1 | 1559517_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7437
p=0.0376
p=0.8301
p=0.0941
p=0.1806
p=0.3408
p=0.5352
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2